



Changes in beer bitterness level during the beer production process

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Abstract

Beer has been enjoyed by consumers for years. Today, hops are inextricably associated with this beverage. Although they have been the subject of research for decades, knowledge of their bittering components and interactions during the beer production process is still incomplete. Current literature clearly indicates that the bitterness experienced in beer comes from a much wider range of compounds than just iso- α -acids. Although compounds that can be classified into β -acids, humulinones, hulupones, hard resins, and polyphenols are characterized by lower levels of bitterness and are present in hops in lower quantities than α -acids, they might determine, together with them, the final level of bitterness in beer. Unlike α -acids, the influence of compounds from these groups, their transformations, changes in their content during the beer production process and factors that affect their final concentration in beer have not yet been thoroughly studied. In case of α -acids, it is known that factors, such as chemical composition of wort, its extract and pH, amount of hops added and α -acids' content, boiling time, and temperature at which hops were added influence the level of bitterness. This phenomenon is further complicated when dry hopping is used. Due to the presence of humulinones, polyphenols, and α -acids, a relatively simple spectrophotometric determination of IBU can give erroneous results. IBU determination, especially in dry-hopped beers, should be coupled with HPLC analysis, taking into account appropriate bitterness coefficients.

Keywords Beer · Hops · Bitterness · Hop resins

Introduction

Beer has been continuously appreciated by the consumers for years. Traditionally, beer is made of three constituents: water, malt, and hop. The last ingredient, hop, has been used for hundreds of years in the beer production process. It imparts a characteristic flavor and aroma to the beer and improves the microbiological stability of the finished product. As a result, nowadays, it is inherently associated with this beverage. The complex chemical composition of raw hop has been a subject of research for decades. Nevertheless, the knowledge about this raw material—its complex interactions and transformations—occurring during the beer production process is still incomplete. It is widely known that

the characteristic beer bitterness is largely due to isomerized α -acids formed by the action of high temperature during boiling of wort with hop. However, this raw material also has a wide range of other compounds that can impart a bitter taste. Available publications indicate that hop fractions so far not recognized as having an impact on beer flavor can significantly alter it. Complex processes occurring at various stages of beer production are no less important than the chemical composition of this raw material. Beside α -acid isomerization during wort boiling, processes such as humulinones extraction or iso- α -acid loss during dry hopping can significantly alter product taste. Awareness of these phenomena can help brewers produce beers with the intended sensory characteristics.

The aim of the paper is to present the current state of knowledge on bitter substances found in hop cones and to describe changes in their concentration in wort during the production process, together with the known factors influencing these transformations.

Contribution for the Special Issue: The chemistry behind malt and beer production – from raw material to product quality.

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Bitter substances in hop cones

Among dozens of flavorings used in brewing over the centuries, only hops (*Humulus lupulus* L.) have become the priority ingredient of this beverage. Around 97% of the world's crop of this perennial plant is intended for brewing purposes. It is a dioecious plant (although it should be noted that there are monoecious plants in wild North American populations). The inflorescences of this plant, commonly known as cones, contain yellow glandular hairs, also known as hop glands or lupulin, which are only found in female plants. It is the presence of these glands that determines the well-established position of hops in the brewing industry [1]. The typical chemical composition of a dried hop cone is shown in Table 1. From the brewing point of view, the most important components of hops are resins and essential oils produced by the mentioned glands. Hop resins usually constitute quantitatively the second fraction in a dry hop cone. Due to their high content of α -acids, they impart the bitterness to the beer. The compounds in this fraction also improve foam stability and microbiological stability of beer. Until now, it was believed that the substances included in hop essential oil fraction only provide aroma; however, there are reports, suggesting that they can increase the perceived level of bitterness and change its character—making it sharper. In terms of sensory attributes, polyphenol content is another important factor, as it can clearly affect beer quality parameters, impart antioxidant properties, and can alter the perceived level of bitterness. McLaughlin et al. [2] conducted a study involving the addition of a specific quantities of polyphenols and/or iso- α -acids to beer. Beers to which polyphenols were added were characterized by higher level of bitterness, which was described as lingering. Additionally, those beers received higher scores in the categories of “metallicity”, “spiciness”, and “medical taste” [2–5].

Table 1 Typical composition of dried hop cone [4]

Compound group	Typical content [%]
Resins	15–30
Essential oils	0.5–3
Proteins	15
Monosaccharides	2
Polyphenols	4
Pectins	2
Amino acids	0.1
Waxes and steroids	trace–25
Ash	8
Moisture	10
Cellulose and other	43

Depending upon the variety and growing conditions, the total hop resin content can be in the range of 15–30% of the dry hop cone weight. This fraction is defined as compounds soluble in diethyl ether or cold methanol. The resins can be divided into soft resins, which are soluble in hexane, and hard resins, which are not (Fig. 1). The extracted soft resins take the form of a thick, viscous liquid resembling in appearance of honey, while the hard resins are a dark dusty solid. Literature sources indicate that the bittering properties of hops are almost exclusively caused by the substances contained in the soft resins fraction, especially α -acids. However, recent publications indicate that bitterness in beer is a much more complex phenomenon, and other, hitherto overlooked compounds can modify its level [3, 4].

α -acids are compounds that are poorly soluble in water. In a study by Baker et al. [7], content of these compounds in hops was in the range of 1.3–12.6 wt%, depending upon variety. However, there are varieties with higher α -acid content. The maximum concentration of these compounds in finished beer is around 14 ppm, but according to Fritsch and Shellhammer [8], α -acids even at concentration of 28 ppm do not impart any perceptible bitterness to beer. When wort is boiled with hops, α -acids are isomerized to the corresponding epimers. It is believed that temperature in excess of 80 °C is required to initiate the reaction. The spacial configuration of the resulting compounds is dependent on the boiling time and temperature. Available literature data indicate that *cis*-iso-humulones are more resistant to high temperatures and aging due to their more stable thermodynamic configuration. Additionally, *cis*-iso- α -acids undergo lower losses during fermentation and storage and are characterized by more intense bitterness than *trans*-iso-humulones. However, they have inferior foam stabilizing abilities to *trans* isomers. For a standard beer wort, the ratio of *cis*:*trans* isomers is typically ~2,5:1. An important indicator concerning the issue of α -acids is their utilization rate. This term is defined as the ratio of iso- α -acids obtained, typically in the finished beer, to the quantity of α -acids that have been added (usually in hops, but they can be added in other forms, such as hop extract). The efficiency of the isomerization reaction is dependent on the number of factors. Various literature sources place typical utilization at less than 30%, or in a range of 40–65%. Verzele and Keukeleire [9] report a maximum utilization during wort boiling to be 60%, while values of 25–30% are found in finished beers, where the content of these compounds decreased due to losses. When using fresh hops, iso- α -acids can be responsible for about 70% of the bitterness of beer, but this value strongly depends on the type of beer. Among these compounds, there are mainly 3 acids: humulone, cohumulone, and adhumulone, which occur in the highest amounts. Other compounds in this group are found only in trace amounts, and for practical reasons, they are usually not mentioned [3, 4, 7–13].

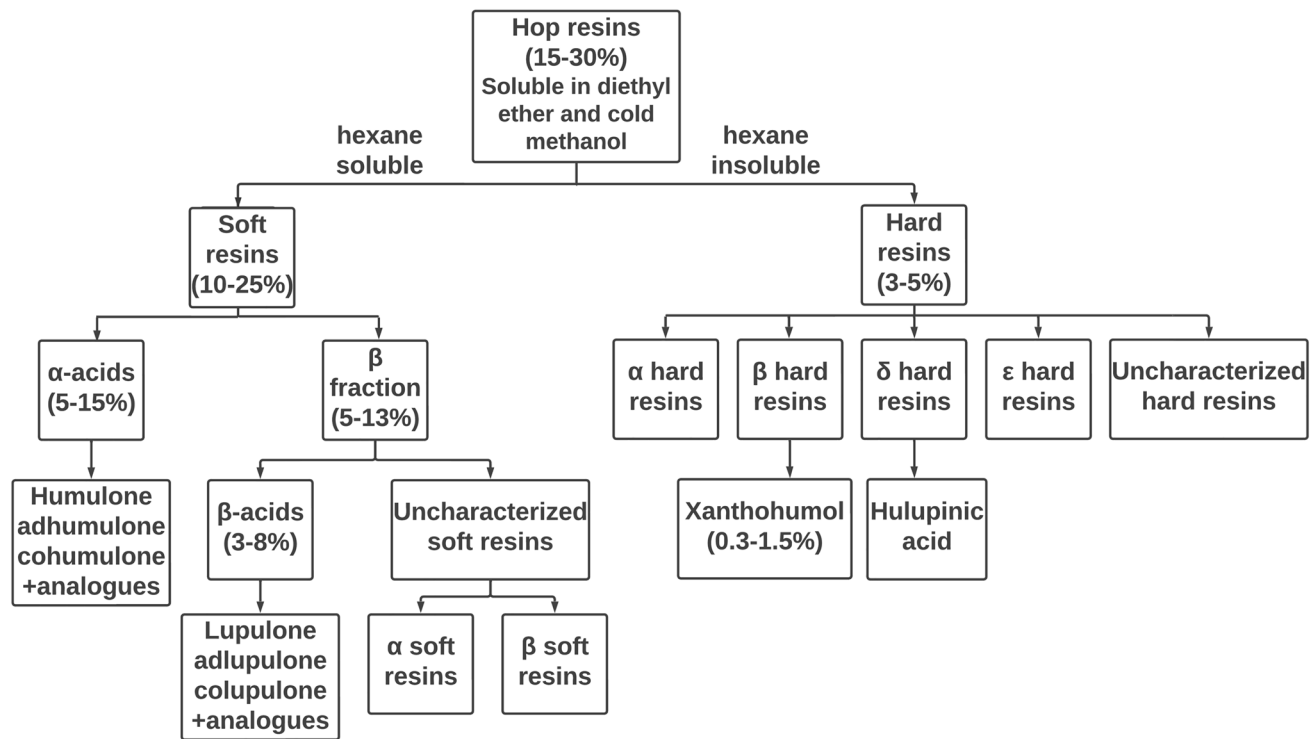


Fig. 1 Hop resins' composition [4, 6]

The β fraction found in soft resins can be divided into β -acids and uncharacterized soft resins. β -acids are characterized by low solubility in water and do not isomerize during boiling. In a study by Baker et al. [7], content of β -acids in the tested hop varieties was in the range of 1.0–6.8 wt%, depending upon variety. Other authors present even broader ranges. The conditions present in the wort (e.g., relatively low pH) cause that only trace amounts of these compounds are transferred into the finished beer. Approximately 70–85% of these compounds remain in hop after boiling; that is why, initially, it was believed that they have no significant effect on beer flavor. However, β -acids show the ability to be oxidized to compounds showing greater solubility in water, which may have a noticeable effect on the sensory characteristics of the beverage. After separation of α - and β -acids, a hitherto uncharacterized fraction of soft resins remains, among which α - and β -fractions are distinguished. So far, the properties of its constituent compounds and potential effects on beer parameters have not been studied. It is believed that compounds in this fraction are intermediate decomposition forms of α - and β -acids, which will eventually become hard resins [4, 6, 7, 14, 15].

Hard resins are defined as the fraction insoluble in hexane. To date, this is the least understood fraction of hop resins. However, current literature indicates that compounds in this group can impart bitterness to beer. Hard resins usually account for 2–3% of dry hop cone weight. In a study

conducted by Almaguer et al. [16], the authors showed that it is possible to obtain beer with desired sensory characteristics using only hard resins. It is believed that a considerable fraction of hard resins are oxidized soft resins. During hop aging, their soft resin content decreases in favor of hard resins. However, it should be noted that hard resins are already detected in the earliest stages of hop cone development, hence primary resins, and those formed by the decomposition of other compounds are distinguished. Almaguer et al. [4] have divided hard resins into five groups: α , β , δ , ϵ , and uncharacterized soft resins. The α fraction represents a small part of the hard resins, and as of today, its brewing properties have not been studied. Among the β -fraction, xanthohumol is quantitatively the most important component of the group, while the other compounds have not been thoroughly characterized. Xanthohumol is a strong antioxidant and exhibits a wide range of antimicrobial and antiparasitic properties. Because of its effects, it is the subject of dozens of studies on its use in, among others, the treatment of metabolic disorders and related diseases. However, it is found in trace amounts in finished beer, because it is isomerized to isoxanthohumol. Both xanthohumol and isoxanthohumol can impart bitterness to beer. However, according to Intelmann et al. (2009), the bitterness level of these compounds is much lower than that of iso- α -acids. The δ -fraction is the only one that is soluble in water, but most likely contributes little to the bitterness in beer. The ϵ -fraction is found in the highest

amounts and can account for up to 80% of the total hard resin content. In a study by Dresel et al. [17], the authors rated the bitterness of 5% alcohol aqueous solutions enriched with hard or soft resins, in the amounts in which they are found naturally in beer, on a 5-point scale (0—bitterness not perceptible, 5—very intense). Soft resins were rated 4.25 (± 0.25), ϵ -resins 0.51 (± 0.08), and total hard resins 0.5 (± 0.25), while the bitterness was nearly undetectable in the δ fraction – 0.08 (± 0.04). These results indicate that most of the bitterness of the hard resins seems to come from the water-insoluble ϵ -fraction.

Humulinones and hulupones are the last distinct fraction found in hop resins. They are intermediate oxidation products of α - and β -acids, respectively. Therefore, they should be classified as hard resins. However, due to their solubility in organic acids, they can also be classified as soft resins. Since it is difficult to clearly classify them into one group or other, they are not listed in the diagram. The typical content of hulupones and humulinones in dry hop cone is usually less than 0.5%. Their content increases with storage time, and they are not detected in fresh hops. In a study by Algazali and Shellhammer [18], the authors estimated the bitterness of humulinones and hulupones to 66 ($\pm 13\%$) and 84% ($\pm 10\%$) of iso- α -acids, respectively. The values found by the authors are higher than previously believed, which indicate that these compounds may have a much more important role than previously thought. Eventually, hulupones are oxidized to un bitter hulupic acid [3, 4, 6, 16–22].

To quantify the scale of bitterness in beer, the International Bitterness Units (IBU) scale was developed in the 1950–60s. At that time, brewers used baled hops that were generally stored under conditions that can be described as unsuitable from today's perspective. Due to inadequate storage conditions, these hops lost 40–80% of their original α -acid content by the time they were used for brewing. This led to the need for quick and simple method to quantify the bittering potential of beer. The determination of IBU involves the extraction of bitter substances from beer with acidified isooctane, followed by the determination of absorbance at $\lambda = 275$ nm. Theoretically, for beers containing only iso- α -acids, 1 IBU $\approx$ 1 mg iso- α -acids/L. Unfortunately, other compounds that show absorbance at the wavelength used, such as α -acids, polyphenols, humulinones, and xanthohumol, also enter the solvent phase. Because they exhibit different levels of absorbance, as well as different levels of bitterness than iso- α -acids, the actual perceived level of bitterness is usually lower than the measured IBU value. This problem becomes particularly apparent in dry-hopped beers. The shortcomings of this method can be avoided with HPLC by determining each component contributing to beers' bitterness separately. However, even with this analysis, factors, such as content of alcohol, polyphenols, and essential oils, together with pH, carbonation level, and water chemistry,

among others, alter the bitterness perceived by the consumer. So far, no internationally accepted method has been established that fully captures the actual level of bitterness experienced [21, 23].

Bitterness level changes during the beer production process

Although hops can be added at any stage of beer production, they are mainly added during boiling and during or after fermentation (in a process known as dry hopping). Hopping during boiling is the traditional method of adding hop, which imparts a desirable level of bitterness. Normally, hops are boiled with wort for 60–90 min. Shorter times (less than 15 min, or even after the boiling is over—while the wort is cooling) are used when hopping for aroma, to retain as much of the highly volatile hop essential oils as possible. Another hopping method, used primarily in India Pale Ale beers, is dry hopping. The main purpose of this method is not the modification of the bitterness levels, but the extraction of volatiles to give the beer eligible, heady scent. Changes in bitterness levels can also occur during beer storage. The following sections describe the changes in levels of bitter compounds that take place during boiling, fermentation, dry hopping, and storage.

Boiling

As mentioned earlier, isomerization of α -acids to iso- α -acids occurs when boiling the wort with hops. Thanks to the isomerization reaction, the un bitter α -acids are transformed to bitter iso- α -acids. The limiting factor for the efficiency of this reaction is the relatively low solubility of these compounds under the conditions found in the wort (approximately 60 mg/L at pH 5.5, 100 °C). Factors affecting the utilization of α -acids include boiling time, wort temperature, sugar content, water chemistry, quantity of hops added, and pH level. Significant losses of α -acids occur due to inadequate boiling times, aerobic conversions, adsorption on solids, precipitation with breakthrough, and losses during fermentation. During the boiling stage, approximately 25–30% of the bittering substances are lost with the spent hops, and another 25–40% in the precipitated breakthrough. Gänz et al. [24] found that the protein substances responsible for the loss of these compounds in the breakthrough are not free amino acids, but uncoagulated proteins. Those that have already precipitated do not react with iso- α -acids. What is interesting, Jaskula et al. [25] report that finishing mashing at 95 °C instead of the common 78 °C allows for higher utilization ($\sim 58\%$ instead of $\sim 42\%$ in the authors' results), and lower concentration of *trans* isomers. The authors contribute the increased utilization to reduced losses of α -acids in the

trub, due to higher protein coagulation at the mashing stage. Those findings are consistent with the data by Gänz et al. [24]. In the study by Jaskula et al. [26], the concentration of polyphenols in the wort had no effect on the loss of bittering substances [9, 11, 12, 24, 25].

In a study by Malowicki and Shellhammer [27], the authors examined the effects of the temperature on the isomerization rate of the α -acids. They evaluated the temperature range of 90–130 °C in a model acetate buffer, pH 5.2, containing on average 64.4 ppm of α -acids. At the standard boiling temperature of 100 °C, 60% isomerization occurred in 90 min. By the time of 60 min which is often used in home brewing, the degree of isomerization was only about 30%. The authors note that in the 100–130 °C temperature range, isomerization occurs more than 2× (229% on average) faster for every 10 °C increase. At 130 °C, 60% isomerization takes place in only 8 min. At the same time, the iso- α -acid degradation products levels increase along with boiling temperature. In the temperature range of 100–130 °C in time required for 60% isomerization, authors found 10.8–14.0% of degradation products. Extending the boiling time beyond the point at which 60% utilization was achieved resulted only in a decrease in the concentration of iso- α -acids, in favor of their degradation products. According to Kappler et al. [11], degradation products can cause an unpleasant, pungent bitterness. In the author's study, a temperature of 90 °C resulted in a more than 2× reduction in the isomerization rate, compared to 100 °C. Similar results were obtained by Jaskula et al. [12]. However, it should be noted that isomerization, although slow, still takes place at 80 °C; hence, keeping the wort after boiling at this temperature should be avoided to prevent the formation of excessive bitterness, or accumulation of degradation products. Hopping for aroma in whirlpool should also be conducted with rapid cooling of the wort [11, 12, 27].

Elevated wort pH can accelerate isomerization of α -acids and reduce their losses. Although the rate of isomerization is not dependent on the pH of the environment (within the range expected in beer wort), α -acids are relatively poorly soluble at low pH levels. Elevated pH increases their solubility which allows for faster reaction pace and higher isomerization rate, by increasing the substrate concentration. In wort with a pH of 8–10, a 90% utilization rate can be achieved. The bitterness obtained in wort with increased pH is described as “unpleasant”. In a study by Jaskula et al. [26], the utilization obtained at pH 6.0 and 7.0 (59.7 ± 1.0 and $81.3 \pm 1.3\%$, respectively) was significantly higher than that at pH 5.2 and 5.5 (37.5 ± 0.7 and $43.4 \pm 0.9\%$, respectively). Kappler et al. [11] examined the recovery rate of preisomerized α -acids after boiling them for 60 min in pH 4–8 buffer. At pH 4.0, only 58% of the original iso- α -acid was found after boil. At pH 5.0, 80% was detected, and at pH 8.0 95%. Reduced utilization

and retention of α -acids may be an important factor in the production of sour beers [11, 26].

Sugar content is another important factor. The prevailing claim in the literature is that high sugar content limits the utilization rate. Different results were obtained by Jaskula et al. [26], where isomerization remained on similar levels in the 10.0–15.2° P extract range. Only in the wort with an extract of 22°P did the degree of isomerization decreased by 11% compared to the 12° P sample. However, non-overlapping results are presented by Kappler et al. [11]. In the authors' study, the extract content affected the preisomerized α -acids concentration. The authors detected that the recovery of preisomerized α -acids decreased linearly from 90% at 10° P, to only 52% at 18° P. The results of Jaskula et al. [26] are also contradicted by the results obtained by Justus [28], where the utilization at an extract of ~20° P was more than twice as low as that obtained at ~12° P. These reports suggests that sugar content does affect the utilization degree [11, 26].

As the hop content increases, the utilization of hop acids decreases. In a study conducted by Irwin et al. [29], the authors observed the highest utilization rate (~30%) of α -acids at their concentration of ~20 mg/L in wort. Utilization decreased with increasing dose of α -acids, reaching value below 20% at their dosage of ~90 mg/L. The authors note that this decrease is not linear. According to Justus [28], the α -acid content of hops also plays a significant role in the utilization. In the study, higher doses of hops with low α -acid content (Tettnang—1.9%) resulted in reduced utilization—30.6%, compared to the analogous in terms of α -acid amount dose of bittering hops (Polaris—17.6%)—45.3% [28, 29].

The last factor that could affect the degree and rate of isomerization is the chemical composition of the water used to produce the beer. The ions found in the greatest quantities in drinking water, and thus also in brewing water are Ca^{2+} , Mg^{2+} , Na^+ , K^+ , SO_4^{2-} , Cl^- , and carbonate ions. Literature indicates that divalent metal ions Ca^{2+} and Mg^{2+} , and monovalent Na^+ and K^+ can accelerate the rate and degree of α -acid utilization. This is confirmed by the study by Jaskula et al. [26], where addition of each of these ions at doses of 5 mg/L resulted in increased utilization rate by 9–23%. Metal ions that are considered as generally undesirable can significantly increase utilization rates. In the author's study, 5 mg/L Fe^{3+} allowed for over 80% utilization rate. The authors used reverse osmosis water for brewing. At the same time, Justus [28] reports that increased Ca^{2+} and Mg^{2+} content can increase breakthrough formation, resulting in turn in reduced utilization. Punčochářová et al. [30] found no statistically significant differences in the content of iso- α -acids in beers brewed with hard and soft water. Due to these ambiguous reports, it is difficult to determine unequivocal role of this ions on beers' bitterness [3, 26, 28, 30].

An interesting issue from a beer storage perspective is the *cis:trans* isomer ratio. Intelmann et al. [31] report that the typical ratio in the wort is 2.5:1. Jaskula et al. [12] report that with shorter than standard boiling times as well as lower wort temperature, the *trans:cis* ratio increases, i.e., more *trans* isomers are formed. This relationship was most pronounced when hop was added for 10 min to the wort at 80 °C, which resulted in nearly 3× greater *trans:cis* ratio than one obtained at 100 °C in time of 90 min. It should be noted that at such low temperatures, isomerization occurs very slowly, and the ratio shifts toward the *cis* isomers with increasing boiling time. Nevertheless, this is another reason to combine whirlpool hopping with intensive cooling. Kappler et al. [11] found that the pH and sugar content of the wort does not affect the ratio of isomers formed [3, 11, 12, 31].

The transformations of β -acids during wort boiling are a much less-understood issue than those of α -acids. As mentioned earlier, these compounds have very low solubility under the conditions found in wort. In a study by Haseleu et al. [14], the authors subjected an isolated colupulone to conditions similar to wort boiling. Among its decomposition products, cohulupone and 5 other compounds were detected, where all of them were characterized by a low sensory threshold and a lingering bitterness. In continuation of this study, Haselau et al. [14] studied transformations of isolated colupulone and lupulone by boiling them in a model wort of pH 5.8. The sensory threshold of obtained compounds depending on specific compound was determined to be in the range of 7.9–90.30 $\mu\text{mol/L}$. Cohulupone produced a bitterness sensation similar to that of iso- α -acids, while tricyclic derivatives gave a sensation of long-lasting, pungent bitterness. The lowest sensory threshold of 7.9 $\mu\text{mol/L}$ was determined for cohulupone, which is lower than the sensory threshold for iso- α -acid *trans*-isocohumulone (19.0 $\mu\text{mol/L}$). The authors also analyzed 12 commercially available beers for the presence of these compounds. Tested beers were characterized by a significant difference in the concentration of these compounds, but in all of the samples, their content was well below their sensory threshold. The sample with the highest cohulupone content, described as a “bitter beer”, contained only $3421 \pm 10.3 \text{ nmol/L}$ of this compound. These publications suggest that β -acid transformation products formed during wort boiling probably do not contribute significantly to bitterness sensation. However, they may exhibit synergistic effects with other bitter substances, which determine the final bitterness of beer [14, 32].

Available literature indicates that compounds present in the hard resin fraction may affect perceived bitterness to a greater extent than β -acids. A study by Dresel et al. [17] showed that the hard resin's bitterness is derived mainly from the ϵ -fraction. In a study, the authors conducted a fermentation trial. When ϵ -resin extract was added after boil,

the bitterness in the beer (27 IBU) was described as medicinal or herbal. When the extract was added at the beginning of boil, the beer (27 IBU) had a pleasant, harmonic bitterness. Almaguer et al. [16] conducted brewing trials in which hops were replaced with extracted hard resins derived from Hallertauer Taurus (bittering hops) and Hallertauer Perle (aroma hops). The resin extract was added at the beginning of the boil (90 min), or 10 min before flameout. Beers in which the resins were added 10 min before the end of the boil showed a very mild level of bitterness. Those in which the resins were added for 90 min showed a mild bitterness that was described as harmonic in both cases, but the beer with bittering hop resins scored better. Additionally, the bitterness was more stable during storage in beers dosed with bittering hop resins. The authors note that hop resins extracted from bittering hops gave overall better beer than those extracted from aroma hops. The obtained results indicate that boiling time changes the perception bestowed to beer by the hard resins [16, 17].

Fermentation

During fermentation, there is a significant decrease in the perceived bitterness. This is caused mainly by the decreasing concentration of iso- α -acids due to a number of factors, among which are: their adsorption on the surface of yeast cells and on surfaces in contact with the beer, their precipitation with sediment, and on the surface of fermentation foam. Additionally, the decrease in pH levels during the fermentation reduces the perceived level of bitterness.

In a study conducted by Justus [28], the average IBU decrease among 14 brewed beers was 33.7%. The authors note that in beers that owe a substantial portion of their bitterness to whirlpool hopping, the decrease in bitterness is greater. This may be due to the formation of higher proportion of *trans* isomers during this type of hopping. It is known that *trans* isomers are more prone to losses during fermentation than *cis* isomers. The use of poorly flocculating yeasts as well as higher wort extract increases the decrease in bitterness. In the authors' study, the greatest decrease in IBU occurred during the first 2 days of fermentation. After first 2–3 days, decreases in IBU were no longer noticeable. In the study by Popescu et al. [33], the average decrease in EBU during fermentation was 27.5–36.19% [28, 33, 34].

Significant changes in IBU levels as well as perceived bitterness occur as a result of dry hopping. The primary purpose of this process is to extract hop oils to impart an aroma to the beer, which is difficult to obtain when hops are added during the boiling. However, current literature clearly indicates that there is a significant change and a split between perceived bitterness and IBU levels thanks to this process. The dry hopping usually results in a decrease in iso- α -acids' levels. Simultaneously, extraction of α -acids,

polyphenols, and humulinones from the hops takes place. It is possible that compounds found in hard resins, especially those from the ϵ - and δ -fractions, may be extracted, affecting the bitterness levels, but so far this issue has not been studied. Due to the lack of available literature, the exact fate and typical content of hulupones is also unknown. B-acids are most likely not extracted during this process due to their poor solubility. Due to the aforementioned different levels of bitterness of individual hop bitter substances, and their different absorbance levels at $\lambda = 275$ nm, the usual IBU analysis may give results far from the actual perceived bitterness.

Maye and Smith [35] investigated changes in bitter compounds levels due to dry hopping. Beers with 51 ppm of iso- α -acids, 0 ppm α -acids, and 0 ppm humulinones were dry-hopped with Cascade hops at 1 lb./bbl. (approximately 3.87 g/L). After 3 days of hopping at 16 °C, HPLC analysis was performed. The iso- α -acid content dropped to 32 ppm, and 13 ppm α -acids and 13 ppm of humulinones were detected in the beer. The IBU determined in the traditional way before dry hopping was 40, and it increased to 49. The authors also conducted an analysis in which the IBU was calculated taking into account the bitterness ratios of the different fractions. In this case, the IBU before hopping was 51, and it dropped to 40.5 after hopping. Comparable results were obtained by Foster et al. [36]. In the authors' study, 4–5% of α -acids and 50–60% of polyphenols present in hops dissolved in beer. As mentioned, α -acids do not have a bitter taste. There is also no literature regarding whether they express some synergistic effects with other compounds in beer, resulting in modifying bitterness levels. However, they show a little absorbance at the wavelength used in the IBU determination, which might interfere with this measurement. Titus et al. [37] investigated the changes in polyphenol concentration during first 1–94 h of dry hopping. Surprisingly, the concentration of polyphenols did not increase gradually with prolonged hopping time, but reached a maximum after 3 h. The polyphenol content of the beer after 24 h of hopping was at its lowest observed level (about 24% less than in the sample after 3 h), after which it gradually increased without exceeding the concentration obtained after 3 h. According to Lafontaine and Shellhammer [38], on average, 75% of humulinones found in hops are extracted into beer at 13.3–15.0 °C in 24 h. With increasing hop amount, the extraction rate decreases to be only 47% at a dosage of 16 g/L. Maye et al. [39] report that in beers with IBU below 25, dry hopping can increase the level of perceived bitterness. In beers above 40 IBU, dry hopping usually decreases this level, unless high hop doses (above 11.6 g/L) are used. The reason for the decrease in perceived bitterness is the decrease in the iso- α -acids' content, which is not sufficiently compensated by the extraction of other bitter substances. Parkin and Shellhammer [40], based on their results, suggest that the increase in bitterness with

dry hopping is mainly due to humulinones, which have a 7–10 $\times$ greater effect on bitterness level than polyphenols. A 7 mg/L increase in humulinone levels increases perceived bitterness analogously to a 15 IBU increase. For a 100 mg/L increase in polyphenols, the increase is only 2.2 IBU. Hahn et al. [41] determined the typical content of polyphenols and humulinones, among 121 commercially available intensively hopped beers. Polyphenol content ranged from 135 to 697 mg/L, with a mean content of 329 mg/L. The average content of humulinones was 17 mg/L. Thus, these compounds may largely account for the perceived bitterness among these beers. Ferreira et al. [21] suggest that humulinones may account for up to 28% of the bitterness in dry-hopped beers. Because hopping during the boil typically uses smaller amounts of hops than dry hopping, and these compounds are found in hops in small amounts, humulinones are only found in significant amounts in dry-hopped beers. The authors determined the losses of these compounds during the production process to be 22% during wort boiling, 14% during fermentation, and 32% during bottle refermentation [21, 35–37, 40–42].

During dry hopping, the pH of beer increases. In a study by Maye et al. [39], the authors dry-hopped a beer with Cascade hops at 0–6 lbs./bbl. The pH increase was nearly linear at 0.1 pH units for every 1 lbs./bbl. (approximately 3.87 g of hops/L). Lafontaine and Shellhammer [38] observed a pH increase of ~ 0.14 at a similar hop dose. The increase in pH due to this process may increase the solubility of α -acids. Additionally, higher pH values result in intensified bitterness sensing. Maye et al. [39] determined the increase in bitterness with a pH increase of 0.1 to be comparable to an increase in IBU of 2–3 units [38, 39].

Beer aging

The biggest problem during beer storage are the quality changes that take place. One such change is the degradation of bittering compounds which progresses with increasing storage time in unsuitable conditions. In the case of beers that have not been dry-hopped, these changes mainly include loss of iso- α -acids. The key factors that determine the changes occurring during storage are temperature and time. In a study by Walters et al. [43], the authors examined the content of iso- α -acids in beer with its original content of 15.3 mg/L, after periods of storage at different temperatures. When the beer was stored for 156 days at 40 °C, their content fell to 4.5 mg/L (71% loss). When stored at 0 and 25 °C for 220 days, there was no change in the concentration of these compounds. The authors noted that *trans*-iso- α -acids, especially at 40 °C, degrade to a much greater extent than *cis*-iso- α -acids. Their minimum concentration was already reached after ~ 120 days of storage. When stored

at 25 °C, the concentration of *cis* isomers remained constant, while *trans* isomers degraded slowly. Their loss was only stopped when beer was stored at 0 °C. Comparable results were obtained by Intelmann [31] and Schmidt et al. [13]. Schmidt et al. [13] note that the variety of hops used had no effects on the loss of *trans* isomers. Intelmann et al. [44] report that compounds derived from the transformation of *trans*-iso- α -acids may be responsible for the formation of an unpleasant, lingering bitterness. In the authors' results, beers with higher pH showed higher stability of these compounds. From the literature data, it appears that when beers hopped only for bitterness are stored at low temperatures, loss of iso- α -acids should not be a major problem. The situation is different in case of beers rich in *trans* isomers. As for now, the only known way to obtain such beer is to apply a whirlpool hopping at 80 °C. As a large number of *trans* isomers form in these conditions, these beers may change their sensory characteristics to a greater extent, in case of improper storage [13, 31, 43, 44].

Since dry-hopped beers contain a different bittering compounds profile (mainly due to the additional significant content of humulinones and polyphenols), they may be more susceptible to improper storage conditions. In a study by Kemp et al. [45], the authors examined the changes that occur when dry-hopped beers are stored for 3, 6, and 10 months at 0, 3, and 20 °C. When stored at 20 °C, the first 3 months brought the greatest decreases in levels of humulinones, α -acids, and iso- α -acids. Degradation of these compounds was minimal in subsequent months. After 10 months of storage, average losses were 22%. When stored at 3 °C, the authors describe the losses as "minimal". Interestingly, after 10 months of storage under these conditions, the levels of α -acids and humulinones increased, at the expense of iso- α -acids loss. In sensory terms, beers stored at 20 °C were considered as less bitter than those stored at 3 °C. Perceived bitterness correlated well with the levels of bitter substances. Ferreira and Collin [46] analyzed Belgian dry-hopped beers after 2 years of storage at 20 °C. The average loss of iso- α -acids under these conditions was 25%, while the average losses of humulones and humulinones were as high as 91% and 73%, respectively. Comparing the results of the authors cited shows a large loss in humulinone content between 10 and 24 months of storage. It is possible that these compounds undergo rapid degradation during this period, or that a yet unknown factor is responsible for their degradation. In a study by Jaskula et al. [12], beers dosed with hop polyphenols extract were characterized by a higher stability of iso- α -acids, which may be related to the antioxidant properties of these compounds. Šibalić et al. [47] report that different hop varieties have quantitatively different composition of polyphenol compounds, which may translate into different antioxidant capacity of the beer, and, as a result, its stability during storage [45–48].

Summary

Current literature indicates that the perceived bitterness in beer may come from a much wider variety of compounds than just iso- α -acids. Although compounds belonging to β -acids, humulinones, hulupones, hard resins, and polyphenols are characterized by lower levels of bitterness, and are found in hops in lower amounts than α -acids, they may determine, along with them, the final level of bitterness in beer. The influence of compounds from these groups, their transformations, and changes in their content during the beer production process along with the factors affecting their final concentration in beer have not yet been thoroughly studied. This phenomenon is further complicated when dry hopping is applied. It also follows that a relatively simple spectrophotometric determination of IBU values, which also detects others compounds with a different bitterness factor, may give erroneous results. Hence, it is closer to the truth to couple IBU analysis with HPLC analysis, taking into account the respective bitterness coefficients. From the moment the wort is obtained, unavoidable changes take place resulting in a deterioration of bitter compounds. Due to the degradation of *trans*-iso- α -acids and humulinones at elevated temperatures, especially beers hopped during whirlpool and dry-hopped should be kept at low temperatures to avoid quality losses.

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Declarations

Conflict of interest The authors declare no competing interests.

Compliance with ethics requirements This article does not contain any studies with human participants or animals performed by any of the authors.

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